

EEG report classification with MedGemma-27B — summary

Turning free-text clinical EEG reports into five structured diagnostic labels (overall abnormality; focal / generalized epileptiform; focal / generalized non-epileptiform) with a small, CPU-run, grammar-constrained **MedGemma-27B**. Benchmarked against the paper’s **Mistral-7B** (Tian et al.) and a **human** second annotator (SG), on 1994 reports (two neurologists, “Zoe” + “Maria”), scored against reference annotator LD.

Key result: our best prompt, **v5**, reaches **87.6%** whole-report accuracy — it **beats Mistral-7B on all five categories** and comes within ~2 points of the human annotator (89.8%).

Where each version lands

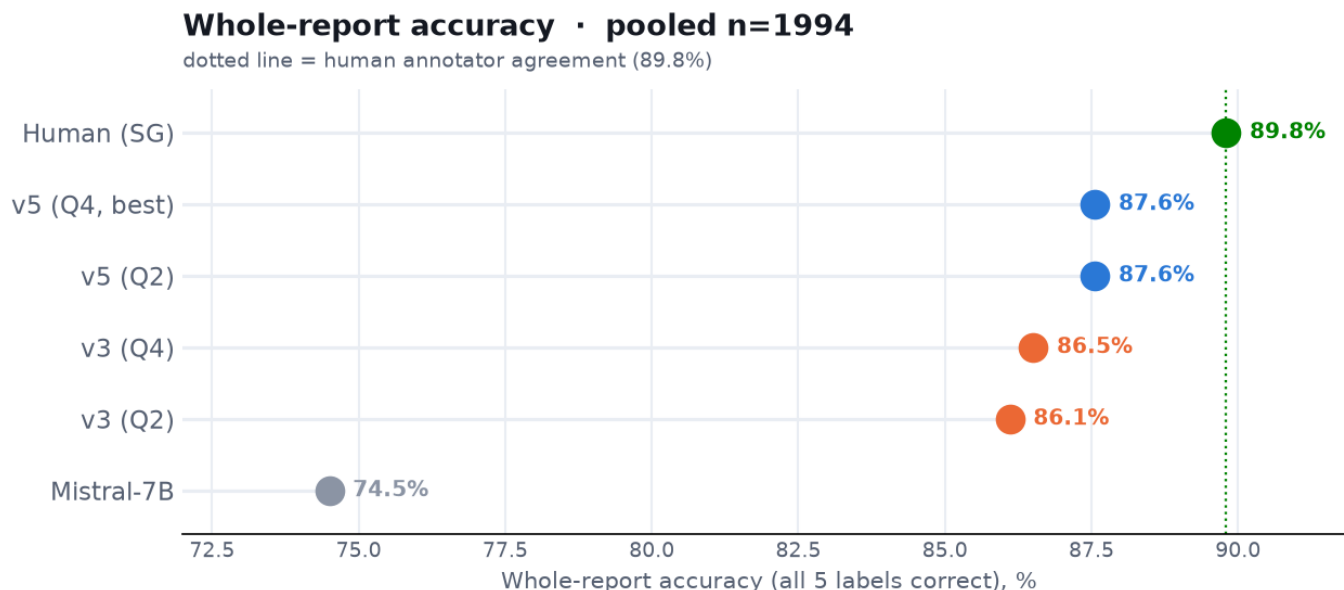


Figure 1: Whole-report accuracy

Two prompt generations, at both model sizes (Q2 ≈ 10 GB, Q4 ≈ 15 GB): **v3** adds an explicit rule for the rare *focal epileptiform* class; **v5** adds a *focal-vs-generalized* rule for slowing. Each step climbs toward the human line; **v5 (Q4)** is the best.

Per category

v5 (blue) beats Mistral (grey) in every category — by 10–15 points on the harder three (Gen Epi, Focal Non-epi, Gen Non-epi) — and lands **at the human level on the epileptiform categories** (Focal Epi, Gen Epi), trailing the human only on Abnormality and focal slowing.

Numbers (pooled n = 1994, Core F1 vs LD)

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	Whole report
Mistral-7B	94.7	82.8	74.8	75.6	75.2	74.5
v3 (Q2)	95.2	88.5	90.4	88.2	85.3	86.1
v3 (Q4)	95.9	86.4	89.2	86.9	89.6	86.5
v5 (Q2)	95.4	87.6	89.9	88.8	86.7	87.6
v5 (Q4, best)	96.0	85.6	88.8	89.0	89.3	87.6
Human (SG)	98.0	85.7	87.5	90.8	90.0	89.8

Per-category F1 — v5 vs v3 vs Mistral-7B vs human

each label scored on its own (Core F1) · v3/v5 at Q4_K_S · pooled n=1994

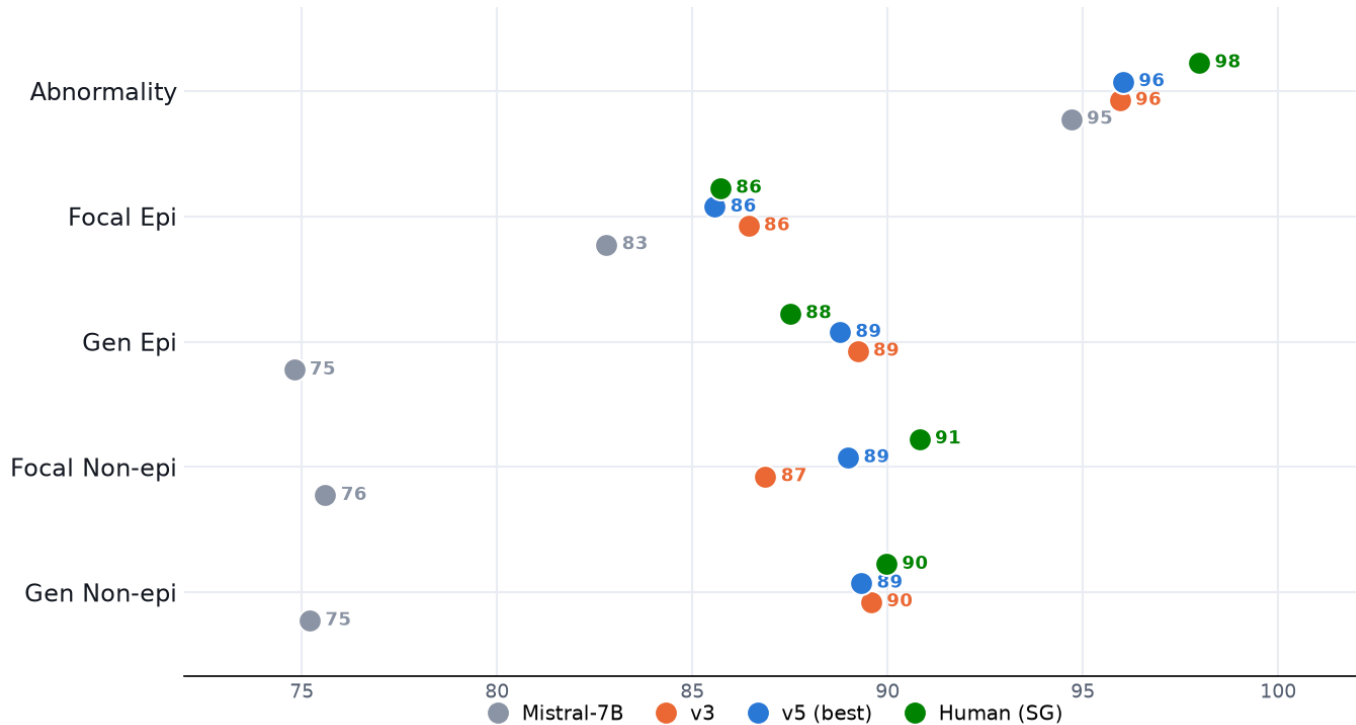


Figure 2: Per-category F1

Per-category values are F1 (0–100), which measures how well each finding is caught — the standard metric for these imbalanced classes. “Whole report” is the share of reports with all five labels correct.

Key points

- **Beats the external baseline everywhere.** v5 is above Mistral-7B in all five categories, decisively on the harder ones (e.g. Gen Epi 89 vs 75, Focal Non-epi 89 vs 76).
- **Near-human.** 87.6% vs the 89.8% human ceiling; on the epileptiform classes it already matches the human.
- **Captures confidence, not just the finding.** When the exact confidence level must match, Mistral collapses on focal epileptiform (F1 0.41) while v5 holds (0.67–0.77).
- **Generalizes.** The result holds on an unseen neurologist (Maria) and against a held-out second annotator — so it reflects real clinical reasoning, not fitting to one reader’s phrasing.
- **Practical.** Runs on CPU; the smaller Q2 model is enough (same whole-report accuracy as Q4, at $\sim\frac{2}{3}$ the size).

v3 and v5 here are run with a consistency-guaranteeing decoding constraint (a grammar that forbids self-contradictory outputs). Full technical detail, all 10 prompt variants, the tested-and-rejected ideas, and per-dataset tables: [all_prompts.md](#) · prompt texts: [../prompts/](#).